

Tennis Prediction Project

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Introduction The aim of this project is testing different machine learning algorithms to predict the outcome of tennis matches and analyzing the results obtained.

Disclaimer After realizing that the average betting market accuracy is around 66%, I tried to build up a model that could beat the market. However, I discovered that regardless the model used, the most important information is embedded in the betting odds, and integrating other match- and player-specific data does not lead to any significant improvement.¹ In this report, I will present the most interesting features that I built to try to reach as much as possible the market accuracy and analyze the results, including the most correct and wrong predictions made by the model.

Datasets and Data cleaning

Two main datasets were used in this project

- **Tennis matches:** A merge of the datasets provided by <http://www.tennis-data.co.uk> with matches from 2005 to 2026 with approximately 55,000 matches.
- **Tennis Players:** A dataset with players' information, including date of birth (dob) and dominant hand.

Initially, the tennis matches started from 2000, but I decided to remove the matches before 2005/07/04, because two important features regarding players' points were not available before that date. After removing duplicates and adjusting feature types, the final dataset is saved in `data/cleaned/tennis_matches.xlsx`.

The tennis players dataset was cleaned as well, removing duplicates, rows with missing values and player names were uniformed to the same format of the tennis matches dataset. (see the examples in Figure 1 and Figure 2). Finally, the dataset was saved in `data/cleaned/tennis_players.xlsx`.

¹Wilkins, 2021, "Sports prediction and betting models in the machine learning age: The case of tennis"

Figure 1: Original player names

	name_first	name_last
0	Jannik	Sinner
1	Novak	Doković
2	Rafael	Nadal

Figure 2: Processed player keys

	player_key
0	sinner j.
1	djokovic n.
2	nadal r.

Feature Engineering

Firstly, I randomly swapped the players in each match to avoid the model learning that the first player is always the winner. Then, I started creating simple features, such as players’ ranking, points and odds differences. I also tested logarithmic differences to handle different ranking tiers, but omitted them from training because tree-based models do not require normalization nor standardization.

Player-specific features I computed the age difference between the two players, restricting ages between 16 and 45 and excluding homonyms. I also defined a left-handed advantage feature to account for left-handers’ rarity and opponents’ lack of practice against them, alongside a home advantage feature, mapping each match location to its corresponding country IOC code to capture crowd support and environmental familiarity.

Player 1	Player 2	Location	Age 1	Age 2	Age Diff	H1	H2	H.Adv	Home 1	Home 2	Home Adv
valkusz m.	sinner j.	Budapest	20.70	17.69	3.01	R	R	0.0	HUN	ITA	-1.0
calatrava a.	djokovic n.	Umag	32.11	18.18	13.94	R	R	0.0	ESP	SRB	0.0
nadal r.	berdych t.	Bastad	19.10	19.81	-0.71	L	R	1.0	ESP	CZE	0.0

Figure 3: Sample matches with player-specific features

ELO This feature estimates a player’s skill level, both overall and surface-specific. Players start with a rating of 1500, with win probability E_1 for Player 1 defined as:

$$E_1 = (1 + 10^{(R_2 - R_1)/400})^{-1}$$

The rating is updated post-match as $\Delta R = K \cdot (S_1 - E_1)$, where R_i are pre-match ratings and $S_1 \in \{0, 1\}$ is the outcome. To account for rating maturity, I employ a dynamic update factor K decreasing with career matches $M_i(t)$ as proposed by FiveThirtyEight²:

$$K_i(t) = 250 \cdot (M_i(t) + 5)^{-0.4}$$

²Bunker et al., 2023, “A comparative evaluation of Elo ratings- and machine learning-based methods for tennis match result prediction”

Surface Elo Computed using the same formula as standard Elo but tracked independently for each surface. To address sparse data on specific courts, it is combined in a 50/50 blend with the overall rating.³ Given its high correlation with standard Elo, only the surface-specific version is retained for model training to avoid redundancy.

Fatigue Player fatigue is modeled using an exponentially weighted moving average (EWMA). It recursively updates the fatigue metric to capture recent workload spikes while giving progressively less weight to older sessions:

$$\text{EWMA}_{\text{today}} = \text{Load}_{\text{today}} \cdot \lambda + (1 - \lambda) \cdot \text{EWMA}_{\text{yesterday}}, \quad \lambda = \frac{2}{N + 1}^4$$

where λ is the degree of decay, N is the fatigue window size in days and $\text{LOAD} \in \{0, 1\}$.

The recursive formula is expanded as:

$$\text{EWMA}_{\text{today}} = \lambda + (1 - \lambda)^{\text{days_gap}} \cdot \text{EWMA}_{\text{last_match}}$$

since on days without matches, $\text{LOAD}_{\text{today}}$ is 0. See the example in Figure 4.

Date	Load	Pre-match Fatigue	Post-match EWMA
Day 0	1 (match)	0 (no history)	$0.2 + 0.8 \cdot 0 = 0.200$
Day 3	1 (match)	$0.8^3 \cdot 0.200 = 0.102$	$0.2 + 0.8 \cdot 0.102 = 0.282$
Day 4	1 (match)	$0.8^1 \cdot 0.282 = 0.226$	$0.2 + 0.8 \cdot 0.226 = 0.381$

Figure 4: Sample fatigue calculation for a single player ($\lambda = 0.2$)

Player form The player recent form is computed as a weighted average of the last N matches won, assigning a higher weight to matches against stronger opponents. The dominance form is calculated using the same logic, considering the games won by the player in the last N matches.

$$\text{weighted_recent_result} = \text{result} \cdot \frac{\text{opponent_ELO}}{\text{BASE_ELO}}$$

$$\text{weighted_dominance_result} = \frac{\text{games_won}}{\text{games_total}} \cdot \frac{\text{opponent_ELO}}{\text{BASE_ELO}}$$

³Jeff Sackmann, 2019, “An Introduction to Tennis Elo”

⁴Murray et al, 2016, “Calculating acute: Chronic workload ratios using exponentially weighted moving averages provides a more sensitive indicator of injury likelihood than rolling averages”

Head to Head statistics They compare two players' historical performance against each other, both overall and surface-specific. See the example in Figure 5.

Match (P_1 vs P_2)	y	Pre-match H2H	Post-match H2H
nadal vs djokovic	1	0	-1
djokovic vs nadal	0	-1	-2
nadal vs djokovic	1	2	-3

Figure 5: Sample head-to-head calculation for two players.

The `pair_key = (djokovic, nadal)` is sorted alphabetically and $y=1$ when P1 wins.

Categorical features Both high and low cardinality categorical features are included in the model and are encoded using one-hot encoding. The high cardinality features, such as **Tournament**, are limited to a maximum of 30 categories and a minimum frequency of 50, grouping the remaining categories into an "other" category.

The prepared dataset is then split into training and testing datasets (70-30).

Models training

The preprocessing pipeline handles both numerical and categorical features:

- **Numerical features:** Missing values are imputed using the median and an indicator column is added to flag missing values.
- **Categorical features:** Missing values are imputed using the most frequent value and one-hot encoding is applied, grouping rare or too frequent categories into "other".

For the classification task, Random Forest and XGBoost were selected. Tree-based algorithms naturally excel in sports prediction by capturing non-linear feature interactions (e.g., fatigue and rankings). Comparing them evaluates Random Forest's robust *bagging* approach against XGBoost's sequential *boosting*, which typically achieves higher predictive accuracy.

Each algorithm is trained with and without betting odds to assess the predictive power of the market and using randomized cross-validation for hyperparameter fine-tuning, employing a time-series split to avoid data leakage.

The average cross-validation accuracy obtained on the training set is about 68% and 70% for the model trained without and with betting odds, respectively.

Results evaluation

Naive baseline The frequency of the most frequent class is about 50%, as we randomly swapped the players in each match. Other interesting baselines regard ranking and points features, which reach an accuracy of about 63.5%. The market odds baseline is 68.51%, which is the most important benchmark to beat.

The results obtained on the test set are summarized in Table 1.

Table 1: Models performance comparison

Model	Accuracy	F1-Score	ROC AUC	Log loss	Brier Score
RF no-odds	65.06%	0.6477	0.7143	0.6169	0.2146
XGB no-odds	65.17%	0.6476	0.7168	0.6156	0.2141
RF with-odds	67.33%	0.6699	0.7397	0.5980	0.2063
XGB with-odds	68.03%	0.6801	0.7471	0.5917	0.2036

The most important features used by the models